Classroom Recording System — Hardware Platform Recommendation

Prepared for: [Institution Name] **Prepared by:** D&R AI Solutions **Date:** April 29, 2026 **Document version:** 1.0

One-Page Summary

The question

We have been running our classroom recording system on Android signage TVs for the past 60+ days. Based on real-world data and the upcoming scale-up to more classrooms, what hardware platform should we use going forward?

The recommendation

Move all new classrooms to Windows Mini PCs. Keep existing Android TVs running until their natural end-of-life.

Why — at a glance

Comparison	Android Signage TV (current)	Windows Mini PC (proposed)
Recording device cost	₹85,000 per room	₹40,000 per room
Recording reliability	~92% successful	~99.5% successful
Camera & mic compatibility	USB only, with limitations	USB + Ethernet + professional AV
Engineering effort to maintain	High (custom systems)	Low (standard Windows tools)
5-year cost for 50 rooms	₹2.52 Cr	₹2.04 Cr
Future-proof	Limited	Excellent

Bottom line

~₹48 lakh saved per 50 classrooms over 5 years. 5× higher reliability. Eliminates ongoing engineering bottleneck.

What we're asking for

1. Approval to port our recording software to Windows (one-time engineering investment, ~₹4 lakh)
2. Approval for a 5-classroom pilot in parallel with existing setup
3. Approval to deploy Windows for all future classroom rollouts

Section 1: Where We Are Today

For the past 60+ days, classroom recording has been running on LG signage TVs with Android built in.

What is working well

- Recordings are being captured and stored reliably to the cloud
- Teachers and students experience smooth playback
- Admin dashboard and reporting is fully functional
- The cloud-side architecture (Azure storage, LiveKit streaming, admin portal) is solid

Where we are running into limits

The hardware itself was designed for displaying digital signage — showing menus, advertisements, dashboards on a screen. It was not designed for recording video for an hour straight.

After 60+ days of pilot operation, three patterns are clear:

1. **The TV's processor overheats during long recordings.** When it does, our system has to temporarily turn off the camera feed to prevent the device from crashing. We've engineered around this, but it's a workaround on inherently underpowered hardware.
2. **USB camera connectivity is unreliable.** Approximately 1–2 disconnect events per 100 hours of recording. Most are silently recovered, but the engineering effort to maintain this is substantial.
3. **The professional camera and microphone you've invested in (Lumens VC-TR1 and Sennheiser TCC2) are designed for Ethernet-based connections.** The Android TV cannot use them at full capability — we are forced to use only the USB pathway, which limits both quality and reliability.

The system works. But we are spending engineering time every week patching the limitations of a hardware platform that wasn't built for this job.

Section 2: The Two Options

Option A — Continue with Android Signage TV

A 55-inch LG commercial TV with Android operating system built in. All-in-one device: it's both the display and the recorder.

Strengths: - Already deployed in 10+ classrooms - Compact, all-in-one form factor - Doubles as a classroom display

Limitations: - Hardware was designed for digital signage, not recording - Processor overheats under sustained load - USB connectivity is the only option (limits camera/mic choice) - Cannot directly use IP-based professional AV equipment - Niche product category — limited manufacturer support - Requires custom-built software update and management system

Option B — Switch to Windows Mini PC

A small box (roughly the size of a hardback book) running Windows 11. Examples: Intel NUC, Dell OptiPlex Micro, HP EliteDesk Mini, ASUS PN-series. Connects to any standard display via HDMI.

Strengths: - Designed specifically for computing tasks - Built-in hardware video encoder (used by professional broadcasters worldwide) - Native support for USB, IP cameras, and audio-over-IP (Dante) -

Excellent manufacturer support (Intel, Dell, HP — Tier-1 vendors) - Standard Windows administration tools — no custom system needed - ~50% lower cost per device

Limitations: - Display is separate (uses your existing TV via HDMI cable) - Windows licensing required (one-time, ~₹8,000–₹15,000 per device at volume) - Initial software migration effort

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Section 3: Cost Comparison

Per-classroom hardware cost

Item	Android TV setup	Windows PC setup
Recording device	₹85,000 (signage TV)	₹40,000 (Mini PC + Windows license)
Display	Included in TV	Use existing display / standard monitor
Camera (Lumens VC-TR1)	₹65,000	₹65,000
Microphone (Sennheiser TCC2)	₹2,80,000	₹2,80,000
Total hardware per room	**₹4,30,000**	**₹3,85,000**
Saving	—	**₹45,000 per room**

Note: The professional camera and microphone costs are identical because both options use the same equipment. The savings come from the recording device itself.

Recurring annual cost (per classroom)

Item	Android TV	Windows PC
Cloud infrastructure	₹2,500	₹2,500
Engineering maintenance	High	Low
Recovery effort from failed recordings	Medium-high	Low
Effective total per year	**~₹15,000**	**~₹2,500**

5-Year Total Cost of Ownership at scale

Deployment size	Android TV total	Windows PC total	You save
50 classrooms	₹2.52 Cr	₹2.04 Cr	**₹48 lakh (19%)**
200 classrooms	₹10.1 Cr	₹8.15 Cr	**₹1.95 Cr (19%)**
500 classrooms	₹25.2 Cr	₹20.4 Cr	**₹4.8 Cr (19%)**

Assumptions: 5-year hardware refresh cycle, ongoing maintenance, cloud infrastructure, and engineering labor at industry-standard rates. Actual numbers depend on volume pricing.

Section 4: Reliability Comparison

Based on 60+ days of pilot data from our deployed Android TVs, compared with industry baselines for Windows-based recording systems:

Metric	Android TV (measured)	Windows PC (industry standard)
Successful recording rate	92%	99.5%
Camera disconnect events per 100 hours	4–6	< 0.5
Thermal throttling per 1-hour class	2–3 events	0
Engineering hours per week to maintain	8–12 hours	1–2 hours

Windows PC numbers reflect deployments at thousands of universities and corporations worldwide running Panopto, Echo360, Mediasite, OBS Studio, Zoom Rooms, etc.

Section 5: Why This Decision Matters Now

Three reasons this is the right moment to decide:

1. The camera and microphone investment

The Lumens VC-TR1 PTZ camera and Sennheiser TCC2 ceiling mic you have already purchased are designed primarily for Ethernet-based deployment. Windows can use them at their full intended quality. The Android TV cannot.

2. We have real data

60+ days of pilot operation has given us concrete evidence — not theory — that the Android TV's limitations are inherent to the hardware, not fixable through more software engineering.

3. Scale is approaching

The decisions made now will impact 50–500 classroom deployments over the next 2–3 years. Locking in the wrong platform multiplies the cost over time.

4. Industry direction

Every major lecture-capture provider — Panopto, Echo360, Mediasite, Zoom — runs on standard PCs (Windows or Linux). None use Android signage TVs. There is a clear reason: signage TVs were built for a different purpose.

Section 6: What Stays the Same, What Changes

What stays exactly the same

- Camera (Lumens VC-TR1)
- Microphone (Sennheiser TCC2)

- Cloud platform (Azure)
- Admin dashboard and reporting
- Teacher and student experience — recording quality, scheduling, playback
- Attendance features and class booking workflow

What changes (behind the scenes)

- The “brain” of the recording moves from inside the TV to a small Mini PC box connected to the TV
- Engineering team uses standard Windows administration tools instead of our custom Android system
- IP-based camera and microphone can be plugged directly into the network — no more USB cable reliability issues

What teachers and students see

Absolutely nothing changes for them. Same recording flow, same playback experience, same dashboard. The change is invisible to end users.

Section 7: Migration Plan

We recommend a careful, phased approach over 12–24 months. Existing classrooms continue to work without disruption.

Phase 1 — Software port (Months 1–2)

- Port the recording engine from Android to Windows
- Validate on a single test classroom
- **Investment:** ~₹4 lakh (engineering time, one-time)
- **Risk:** Low — Windows is well-understood territory

Phase 2 — Pilot deployment (Months 3–4)

- Deploy 5 Windows Mini PCs in new classrooms (or as replacements for failing units)
- Run side-by-side with existing Android TVs for 30 days
- Validate reliability claims with real numbers
- **Investment:** ~₹20 lakh hardware
- **Risk:** Very low — parallel operation, no service disruption

Phase 3 — Production rollout (Months 5–12)

- All new classroom rollouts use Windows
- Existing Android TVs continue operating
- Procurement and deployment standardize on the new platform
- **Investment:** ₹3.85 lakh per new classroom

Phase 4 — Natural transition (Years 2–4)

- As Android TVs reach end-of-life (typically 3–4 years), they get replaced with Windows units
- No need to retire working hardware prematurely

- Eventually all classrooms are on the unified Windows platform

Key point: No existing classroom is disrupted. The transition happens naturally, at the pace of normal hardware replacement.

Section 8: Risks and Mitigations

Risk	Likelihood	Impact	How we handle it
Software port takes longer than 2 months	Low	Medium	Use industry-standard libraries; we control timeline
Windows licensing cost adds up	Low	Low	Volume pricing is well-understood; still cost-positive at any scale
Existing Android TV support gradually weakens	Medium	Low	Continue current support; new deployments avoid the platform anyway
Mini PC hardware vendor issues	Low	Low	Multiple vendor options (Intel, Dell, HP, ASUS) all available
Staff training required for Windows admin	Low	Low	Standard Windows skills are widely available; less specialized than Android signage

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Section 9: Decision Points for Your Approval

1. ☐ **Approve continued operation of existing 10+ Android TVs** No action required — they keep working. Supported until natural replacement.
2. ☐ **Approve software port to Windows (~₹4 lakh, one-time)** Unlocks all the cost savings and reliability gains below.
3. ☐ **Approve 5-classroom pilot deployment** Validates the reliability claims with real numbers before committing to scale.
4. ☐ **Approve Windows as the platform for all future classroom deployments** Locks in ~₹48 lakh savings per 50 classrooms; future-proofs the system.

Section 10: Frequently Asked Questions

Q: Will recording quality stay the same? A: Yes — and slightly better, because Windows can use the professional IP-based connections that the Android TV cannot.

Q: Will teachers and students notice the change? A: No. The user experience is identical.

Q: What do we lose by switching to Windows? A: The Mini PC is a separate box from the display, rather than all-in-one. That's the trade-off for ~₹45,000 hardware savings, 5× higher reliability, and future-proofing.

Q: What if a Mini PC fails? A: Replacement is fast (any IT vendor stocks them) and cheap (₹40,000 vs ₹85,000). Spare-parts strategy is much simpler than for niche signage TVs.

Q: Why didn't we choose Windows from the start? A: At kickoff, the all-in-one TV form factor was attractive. After 60+ days of real-world data, we now have evidence that the trade-offs aren't worth it. Decisions improve with data.

Q: What about Linux instead of Windows? A: Linux is technically capable but harder for typical IT teams to support. Windows is the standard for commercial AV deployments — staff with Windows skills are easier to find and train.

Q: Can we keep both running together? A: Yes. Existing 10 Android TVs continue. New classrooms use Windows. They share the same backend cleanly — no parallel infrastructure needed.

Q: When do we see the savings start? A: Per-classroom cost savings begin from the very first Windows deployment in Phase 2. Reliability gains are measurable within 30 days of pilot.

Q: Is the Mini PC powerful enough? A: Yes. Even entry-level Mini PCs (~₹40,000) have several times the recording capacity of the current Android TV. Professional broadcasters use the same class of hardware.

Section 11: Sign-off

Decision	Approved by	Date
Continue current Android TV operations	_____	___
Approve Windows software port (₹4L)	_____	___
Approve 5-room pilot	_____	___
Approve Windows for new deployments	_____	___

Prepared by: D&R AI Solutions Engineering Team Contact: Dibyakanta Acharya, Founder & CEO

Document basis: This recommendation is based on 60+ days of pilot operation across multiple classrooms, direct engineering analysis of both platforms, industry benchmarking against major lecture-capture providers, and Total Cost of Ownership modelling at 50/200/500 classroom scale.

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